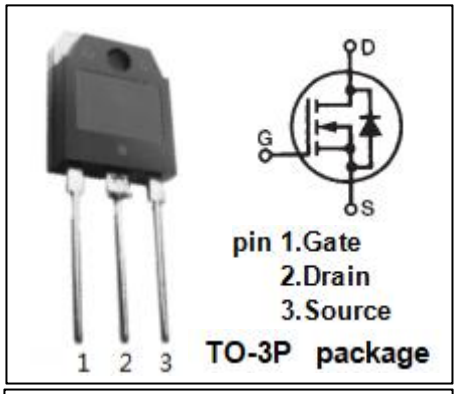
isc N-Channel MOSFET Transistor MTH7N45



FEATURES

·Drain Current -I_D= 7A@ T_C=25℃

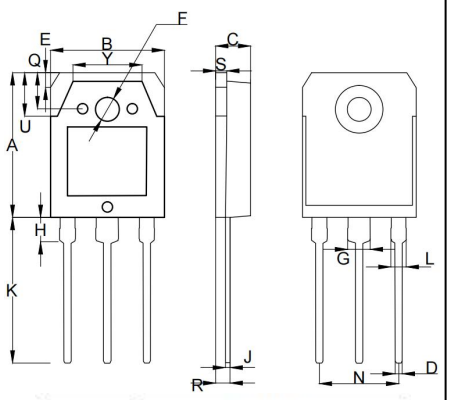
·Drain Source Voltage -V_{DSS}= 450V(Min)

·Static Drain-Source On-Resistance

-$R_{DS(on)}$ = 0.8Ω(Max)@V_{GS}= 10V

DESCRIPTION

• Motor drive, DC-DC converter, power switch and solenoid drive.



DIM	mm		
	MIN	TYP.	MAX
A	19.70	19.95	20.20
B	15.45	15.60	15.75
C	4.70	4.80	4.90
D	0.90	1.00	1.10
E	1.80	2.00	2.20
F	3.40	3.50	3.60
G	2.95	3.10	3.25
H	3.10	3.25	3.40
J	0.50	0.60	0.70
K	19.80	20.00	20.20
L	2.00	2.10	2.20
M	13.30	13.60	13.90
N	10.80	10.90	11.00
Q	4.25	4.68	5.10
R	1.75	2.03	2.30
S	1.40	1.50	1.60
U	5.90	6.05	6.20
Y	9.45	9.60	9.75

ABSOLUTE MAXIMUM RATINGS(T_a=25℃)

SYMBOL	PARAMETER	VALUE	UNIT
V_{DSS}	Drain-Source Voltage	450	V
V_{GS}	Gate-Source Voltage-Continuous	±20	V
I_D	Drain Current-Continuous	7	A
I_{DM}	Drain Current-Single Pluse	40	A
P_D	Total Dissipation @T_C=25℃	150	W
T_J	Max. Operating Junction Temperature	150	℃
T_{stg}	Storage Temperature	-55~150	℃

THERMAL CHARACTERISTICS

SYMBOL	PARAMETER	MAX	UNIT
$R_{th\ j\text{-}c}$	Thermal Resistance, Junction to Case	0.83	℃/W

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ELECTRICAL CHARACTERISTICS（T_C=25℃ unless otherwise specified）

SYMBOL	PARAMETER	CONDITIONS	MIN	MAX	UNIT
$V_{(BR)DSS}$	Drain-Source Breakdown Voltage	V_{GS}= 0; I_D= 0.25mA	450	--	V
$V_{GS(th)}$	Gate Threshold Voltage	V_{DS}= V_{GS}; I_D= 1mA	2.0	4.5	V
$R_{DS(on)}$	Drain-Source On-Resistance	V_{GS}= 10V; I_D= 3.5A	--	0.8	Ω
I_{GSS}	Gate-Body Leakage Current	V_{GS}= ±20V;V_{DS}= 0	--	±100	nA
I_{DSS}	Zero Gate Voltage Drain Current	V_{DS}= 450V; V_{GS}= 0	--	200	uA
V_{SD}	Forward On-Voltage	I_S= 7A; V_{GS}= 0	--	1.8	V

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